Elliptic and Parabolic Partial Differential Equations and Finite Difference Methods

Differential problems, 1D example (not a PDE, but an ODE)

Find $u : [a, b] \to \mathbb{R}$ (with $u \in C^2([a, b])$) such that

$$-u''(x) = f(x)$$

Let's take $f(x) \equiv 1$.

The solutions of the equations are all the parabolas with -1 as coefficient of the quadratic term. To select a unique solution we need to impose some boundary conditions, for example $u(0) = u(1) = 0$. The unique solution of the problem is then

$$u(x) = \frac{1}{2}x(1 - x).$$

Let's generalize to more dimensions!

Poisson equation

Given a domain $\Omega \in \mathbb{R}^d$ where $d > 1$, we seek for $u : \Omega \to \mathbb{R}$ solution of

$$-\Delta u(x) = f(x) \text{ for } x \in \Omega,$$

where Δ is the Laplacian operator, i.e., $\partial^2_{x_1} + \partial^2_{x_2} + \cdots + \partial^2_{x_d}$.

We say the equation is homogeneous if $f \equiv 0$. The homogeneous Poisson equation is called Laplace equation.

Physical applications

- Heat conduction (u temperature),
- Elastic membrane subject to a body force f (u is the displacement),
- Electric potential distribution (u) due to a charge f.

Physical derivation

u is a concentration, at equilibrium there will be zero net flux $\mathbf{F}$ through the boundary of any regular subdomain $S \subset \Omega$, so

$$\int_{\partial S} \mathbf{F} \cdot \mathbf{n} \mathrm{d}\Gamma = 0,$$

using the divergence theorem, we have that

$$\int_S \mathrm{div} \mathbf{F} \mathrm{d}x = \int_{\partial S} \mathbf{F} \cdot \mathbf{n} \mathrm{d}\Gamma = 0,$$

so $\mathrm{div}\mathbf{F} = 0$. In many applications it is reasonable to assume that the flux is proportional to the gradient of the concentration (chemical concentration, heat), from the higher concentration region to the lower ones $\mathbf{F} = -a\nabla u$.

Substituting we get

$$\mathrm{div}\mathbf{F} = -\mathrm{div}(a\nabla u) = -a\Delta u = 0.$$

Boundary conditions

To obtain uniqueness of the solution, we need to enforce some extra constraints.

Dirichlet boundary conditions

$$u = g \text{ on } \partial\Omega.$$

If $g \equiv 0$, then they are called homogeneous Dirichlet BC.

Heat application -> thermal bath at fixed temperature.

Neumann boundary conditions

$$\nabla u \cdot \mathbf{n} = \frac{\partial u}{\partial \mathbf{n}} = h \text{ on } \partial\Omega.$$

$\mathbf{n}$ is the normal vector going out of the domain Ω in each point of the boundary $\partial\Omega$.

Heat application -> prescribed heat flux or temperature change rate on the boundary (e.g. no heat flux is an isolated boundary).

Existence and uniqueness of the solution

Given the Poisson problem with a bit of Dirichlet BC and $f \in L^2(\Omega)$, we have existence and uniqueness of the solution.

Heat equation

Given a domain $\Omega \in \mathbb{R}^d$ we look for a solution $u : \Omega \times \mathbb{R}^+ \to \mathbb{R}$ solution of

$$\partial_t u(t,x) - a\Delta u(t,x) = f(t,x,u), \qquad \partial_t u - a\partial_{xx} u = f \ (\text{in 1D} + \text{time}),$$

with $a > 0$.

Physical applications

- Heat conduction (u temperature, a thermal conductivity, u_0 initial temperature, Dirichlet = Thermal bath, Neumann = temperature change rate),
- Elastic membrane subject to a body force f (u is the displacement),
- Electric potential distribution (u) due to a charge f.

Difference with Elliptic

- Variation in time

Cauchy problem

We couple the PDE with initial conditions (IC) at time $t = 0$ AND boundary conditions (either Neumann or Dirichlet) for all times $t \in \mathbb{R}^+$.

$$\begin{cases} \partial_t u(t,x) - a\partial_{xx} u(t,x) = f(t,x,u), & t > 0, x \in \Omega \\ u(0,x) = u_0(x), & x \in \Omega, \\ u(t,x) = u_D(t,x), & \forall t \in \mathbb{R}^+, x \in \Gamma_D \subset \partial\Omega, \\ \partial_x u(t,x) \cdot \mathbf{n} = u_N(t,x), & \forall t \in \mathbb{R}^+, x \in \Gamma_N \subset \partial\Omega. \end{cases}$$

Today's Working Problems ($d = 1$ or $d = 2$)

Poisson equation

$$\begin{cases} \Omega = [0,1] \\ -\Delta u(x) = f(x) \text{ in } \Omega := [0,1]^d \\ u(x) = 0, \quad \forall x \in \partial\Omega. \end{cases}$$

Heat equation

$$\begin{cases} \Omega = [0,1] \\ \partial_t u(t,x) - a\Delta u(t,x) = f(t,x), & t > 0, x \in \Omega := [0,1]^d \\ u(0,x) = u_0(x), & x \in \Omega, \\ u(t,0) = u(t,1) = 0, & \forall t \in \mathbb{R}^+, x \in \partial\Omega. \end{cases}$$

Allen-Cahn

$$\begin{cases} \Omega = [0,1] \\ \partial_t u(t,x) - a\Delta u(t,x) = r(u(x) - u^3(x)), & t > 0, x \in \Omega := [0,1]^d,\ a, r > 0, \\ u(0,x) = u_0(x), & x \in [0,1]^d, \\ u(t,0) = u(t,1) = 0, & \forall t \in \mathbb{R}^+, x \in \partial\Omega. \end{cases}$$

Finite Difference for Differential Equations

We have seen how the equations are defined. In some cases there are exact solutions, but in many other cases there is no known analytical solution.

Numerical analysis comes in help to find **approximations** of such solutions.

We start with the simplest approach one can think of:

Finite Difference

Finite Difference

Consider a boundary value 1D problem on $\Omega = [a, b]$ defined as

$$F(x, u, \partial_x u, \partial_x^2 u, \ldots, \partial_x^k u) = 0 + BCs.$$

Since the computer represents only discrete states and we want to approximate a function u, we can do it on a simple grid:

$$a = x_0 < x_1 < \cdots < x_i < \ldots x_N = b$$

with values

$$u_i \approx u(x_i).$$

To transform the PDE into a system of $N + 1$ equations for our $N + 1$ unknown, we have to decide how to deal with the derivatives in a discrete sense.

For simplicity, we will consider regular grids with $x_{i+1} - x_i = h$ for all $i = 0, \ldots, N - 1$.

Let's start with the first derivative $\partial_x u(x)$!

Definition: divided difference for first derivative

Let $u : [a, b] \to \mathbb{R}$ be a bounded function, let $x \in [a, b]$ and $h \in \mathbb{R}^+$. We define the **forward divided difference** of f in x with spacing h by

$$\delta_{h,+} u(x) = \frac{u(x+h) - u(x)}{h},$$

for $x + h \in [a, b]$ and, we define the **backward divided difference** of f in x with spacing h by

$$\delta_{h,-} u(x) = \frac{u(x) - u(x-h)}{h},$$

for $x - h \in [a, b]$ and the **central divided difference** of f in x with spacing h by

$$\delta_h u(x) = \frac{u(x+h/2) - u(x-h/2)}{h},$$

for $x + h/2, x - h/2 \in [a, b]$.

Central divided difference for second order derivative

We apply recursively the left and the right first order divided differnce

$$\delta_h^2 u(x) := \delta_{h,-}\left(\delta_{h,+}u\right)(x) = \delta_{h,-}\left(\frac{u(x+h)-u(x)}{h}\right) = \frac{u(x+h)-2u(x)+u(x-h)}{h^2}.$$

If u is has the second derivative $u''(x)$ in x, then

$$\lim_{h\to 0}\delta_h^2 u(x) = u''(x).$$

Taylor expansion!

Suppose that u is regular enough, let's expand the divided differences in x and see what we get.

$$\delta_h^2 u(x) - u''(x) = \frac{u(x+h)-2u(x)+u(x-h)}{h^2} - u''(h) =$$

$$\frac{u(x)+hu'(x)+\frac{h^2}{2}u''(x)+\frac{h^3}{6}u'''(x)+\frac{h^4}{24}u^{(4)}(x)-2u(x)u(x)-hu'(x)+\frac{h^2}{2}u''(x)-\frac{h^3}{6}u'''(x)+\frac{h^4}{24}u^{(4)}(x)}{h^2} - u''(x)$$

$$= \frac{h^2}{12}u^{(4)}(\xi)$$

$$|\delta_h^2 u(x) - u''(x)| \leq \frac{h^2}{12}\max_{\xi\in[a,b]}|u^{(4)}(\xi)|.$$

Divided differences for two-point boundary problems

Let's start with a simple boundary value problem:
find $u : [a, b] \to \mathbb{R}$ such that

$$-u''(x) = f(x) \text{ for } x \in [a, b]$$

and $u(a) = \alpha,\ u(b) = \beta$, where $f : [a, b] \to \mathbb{R}$ is a given function. Suppose that we cannot find an analytical solution, how can we approximate it?

We try to approximate the solution in some equispaced points x_i with $i = 0, \ldots, N$ with $x_{i+1} = x_i + h$, $x_0 = a$ and $x_N = b$.

We have seen that we can approximate the derivatives with divided differences, for **second order** we use the **central divided difference**, so we get

$$-u''(x_i) \approx \frac{u(x_{i+1}) - 2u(x_i) + u(x_{i-1})}{h^2} \qquad \forall i = 0, \ldots, N.$$

So, we can look at an approximation of $u(x_i) \approx u_i$ for all $i = 0, \ldots, N$ that solves the following system

$$-\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} = f(x_i) \qquad \forall i = 1, \ldots, N-1.$$

Matrix form

$$-\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} = f(x_i) \qquad \forall i = 1, \dots, N-1.$$

We can write this system in matrix form as

$$-\frac{1}{h^2}\underbrace{\begin{bmatrix} -2 & 1 & 0 & \dots & 0 \\ 1 & -2 & 1 & \dots & 0 \\ 0 & 1 & -2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & -2 \end{bmatrix}}_{=:D_2}\underbrace{\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_{N-1} \end{bmatrix}}_{=:\mathbf{u}} = \underbrace{\begin{bmatrix} f(x_1) \\ f(x_2) \\ f(x_3) \\ \vdots \\ f(x_{N-1}) \end{bmatrix}}_{=:F}$$

We can use our favourite linear solver to find the solution U from $A\mathbf{u} = F$, with $A = -\frac{1}{h^2}$.

$N-1$ equations for $N+1$ unknowns! We have forgotten u_0 and u_N.

Dirichlet Boundary Conditions

We want to impose that $u(a) = u_0 = \alpha$ and $u(b) = u_N = \beta$. How can we modify the system?
Clearly the first equation for u_1 becomes

$$-\frac{u_0 - 2u_1 + u_2}{h^2} = f(x_1) \Longrightarrow -\frac{\alpha - 2u_1 + u_2}{h^2} = f(x_1).$$

But how can we incorporate this into the system without interfering with the other equations?
Add an artificial equation for u_0 and u_N: $u_0 = \alpha \text{ and } u_N = \beta$.
Then, we can add the equations for u_0 and u_N to the system $A\mathbf{u} = F$ and solve the system as before.

$$\underbrace{-\frac{1}{h^2}\begin{bmatrix} -h^2 & 0 & 0 & 0 & \dots & 0 & 0 \\ 1 & -2 & 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & -2 & 1 & \dots & 0 & 0 \\ 0 & 0 & 1 & -2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & -2 & 1 \\ 0 & 0 & 0 & 0 & \dots & 0 & -h^2 \end{bmatrix}}_{=:A} \underbrace{\begin{bmatrix} u_0 \\ u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_{N-1} \\ u_N \end{bmatrix}}_{=:\mathbf{u}} = \underbrace{\begin{bmatrix} \alpha \\ f(x_1) \\ f(x_2) \\ f(x_3) \\ \vdots \\ f(x_{N-1}) \\ \beta \end{bmatrix}}_{=:F}$$

```
e = ones(N,1); A=spdiags([-1*e 2*e -1*e],[-1 0 1],N,N)/h^2; A(1,:)=0; A(1,1)=1; A(N,:)=0; A(N,N)=1;
```

Neumann Boundary Conditions

Suppose now that the BC are of **Neumann** type on the **left**, i.e., $\partial_x u(a) = \alpha$ and $u(b) = \beta$. How can we modify the system?
We have to approximate at the border the derivative with a divided difference.

$$\partial_x u(a) \approx \frac{u_1 - u_0}{h} = \alpha.$$

Exercise

Modify the linear system to include the Neumann BC on the left and the Dirichlet BC on the right.

Exercise

Find the finite difference approximation of the homogeneous Dirichlet Poisson problem on a non uniform grid, i.e., $x_{i+1} - x_i \neq x_i - x_{i-1}$.

Existence and uniqueness of the solution?

Let's just focus on the matrix A given by the 1D Poisson problem with Dirichlet BCs (or mixed Dirichlet-Neumann).
We can see that the matrix (excluding the Dirichlet BC) is

- symmetric
- diagonally dominant
- diagonal is positive
- hence, it is positive definite ($\implies$ coercive!)
- This implies that the matrix is invertible and the solution exists and is unique.

Finite Difference Discretization of $\partial_t u - \partial_{xx} u = 0$

- Given a spatial discretization of $\partial_x x$ we are left with $\partial_t \mathbf{u} = A\mathbf{u}$, with $A \approx -\partial_{xx}$.
- Domain in space $\Omega = [a, b]$ and time $[0, T]$
- Grid in space $a = x_0 < x_1 < \cdots < x_i < \cdots < x_{N_x} = b$
- Grid in time $0 = t^0 < t^1 < \cdots < t^n < \cdots < t^{N_t} = T$
- $u_i^n \approx u(t^n, x_i)$

Explicit Euler

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} - \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2} = 0$$

Implicit Euler

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} - \frac{u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}}{\Delta x^2} = 0$$

Crank-Nicolson

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} - \frac{u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}}{2\Delta x^2} - \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{2\Delta x^2} = 0$$

Numerical solutions

Explicit Euler

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} - \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2} = 0$$

- Explicit -> no systems

Implicit Euler

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} - \frac{u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}}{\Delta x^2} = 0$$

- Linear system

$$LHS = I - \frac{\Delta t}{\Delta x^2} D_2 = \begin{pmatrix} 1 + 2\frac{\Delta t}{\Delta x^2} & -\frac{\Delta t}{\Delta x^2} & 0 & \dots & \dots \\ -\frac{\Delta t}{\Delta x^2} & 1 + 2\frac{\Delta t}{\Delta x^2} & -\frac{\Delta t}{\Delta x^2} & \dots & \dots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & -\frac{\Delta t}{\Delta x^2} & 1 + 2\frac{\Delta t}{\Delta x^2} \end{pmatrix} \qquad RHS = u^n$$

Consistency

Explicit Euler

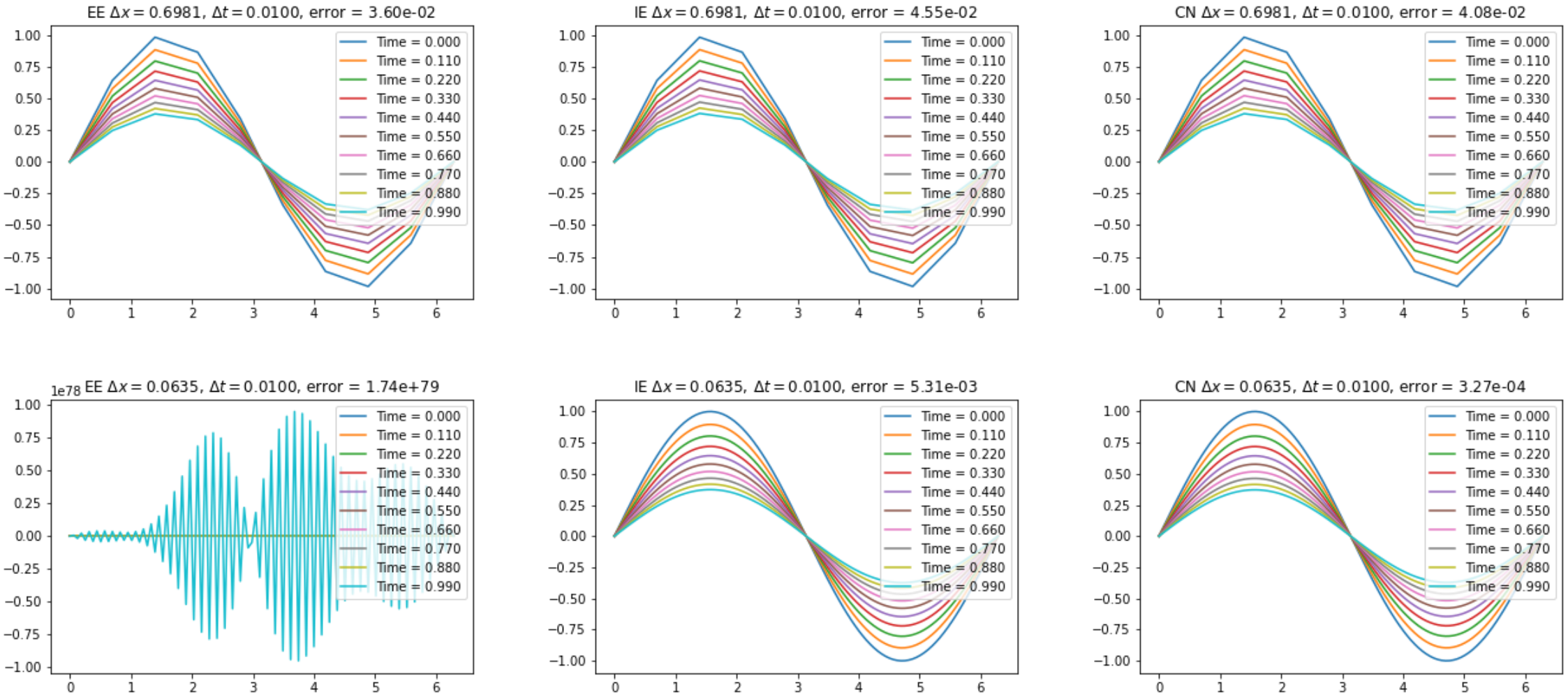
$$\frac{u_i^{n+1} - u_i^n}{\Delta t} - \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2} = 0$$

$$\begin{aligned}
e_{\Delta t, \Delta x}^{EE} =& \frac{u(t^{n+1}, x_i) - u(t^n, x_i)}{\Delta t} - \frac{u(t^n, x_{i+1}) - 2u(t^n, x_i) + u(t^n, x_{i-1})}{\Delta x^2} \\
=& \partial_t u(t^n, x_i) + \frac{\Delta t}{2} \partial_{tt} u(t^n, x_i) - \partial_{xx} u(t^n, x_i) - \frac{\Delta x^2}{12} \partial_{xxxx} u(t^n, x_i) + O(\Delta t^2) + O(\Delta x^3) \\
=& \frac{\Delta t}{2} \partial_{tt} u(t^n, x_i) - \frac{\Delta x^2}{12} \partial_{xxxx} u(t^n, x_i) + O(\Delta t^2) + O(\Delta x^3) = O(\Delta t) + O(\Delta x^2)
\end{aligned}$$

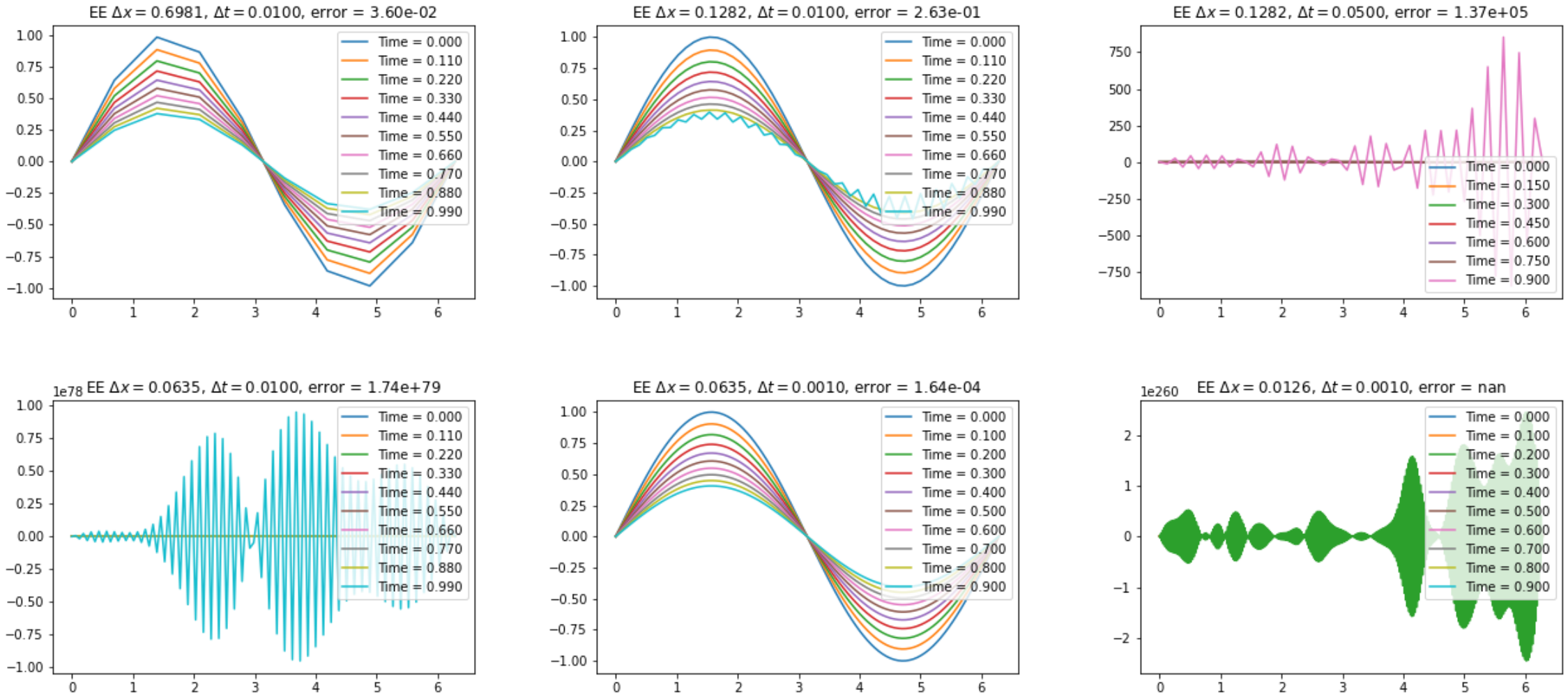
Second order in space and first order in time

Example

$$\begin{cases} \partial_t u - \partial_{xx} u = 0, \\ u_0(x) = \sin(x) \qquad x \in [0, 2\pi], \\ u(t,0) = u(t,2\pi) = 0. \qquad t \in \mathbb{R}^+, \end{cases} \qquad u(t,x) = e^{-t}\sin(x) \qquad x \in [0,2\pi], \qquad t \in \mathbb{R}^+.$$



Explicit Euler: Stability issues $\Delta t < C\Delta x^2$



Nonlinear problems (Allen-Cahn)

$$\partial_t u - \partial_{xx} u = u - u^3.$$

Spatial semidiscretization

$$\partial_t \mathbf{u} = A\mathbf{u} + F(\mathbf{u}),$$

where $A \approx \partial_{xx}$ and $F(\mathbf{u}) = \mathbf{u} - \mathbf{u}^3$.

Implicit Euler

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} - \frac{u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}}{\Delta x^2} = u_i^{n+1} - (u_i^{n+1})^3$$

Stable, but nonlinear!

Nonlinear solver at each time step, e.g. Newton's method

$$\text{Find } \mathbf{u}^{n+1} \text{ such that } R(\mathbf{u}^{n+1}) = 0, \text{ where } R : \mathbb{R}^{N_x} \to \mathbb{R}^{N_x},$$

$$R(u^{n+1})_i = \frac{u_i^{n+1} - u_i^n}{\Delta t} - \frac{u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}}{\Delta x^2} - u_i^{n+1} + (u_i^{n+1})^3$$

Newton's method for nonlinear problems (1/2)

$$R(u^{n+1})_i = u_i^{n+1} - u_i^n + \Delta t \left(-\frac{u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}}{\Delta x^2} - u_i^{n+1} + (u_i^{n+1})^3 \right)$$

$$J_{u^{n+1}} R(u^{n+1})_{ij} = \partial_{u_j^{n+1}} R(u^{n+1})_i = \delta_{ij} + \Delta t \left(-\frac{\delta_{i+1,j} - 2\delta_{i,j} + \delta_{i-1,j}}{\Delta x^2} - \delta_{ij} + 3(u_i^{n+1})^2 \delta_{ij} \right)$$

$$J_{u^{n+1}} R(u) = I + \Delta t \left(-\frac{D_2}{\Delta x^2} - I + 3\mathrm{diag}(u^2) I \right) =$$

$$\begin{pmatrix} 1 + \Delta t(2\frac{1}{\Delta x^2} - 1 + 3(u_1)^2) & -\frac{\Delta t}{\Delta x^2} & 0 & \dots & \dots \\ -\frac{\Delta t}{\Delta x^2} & 1 + \Delta t\left(2\frac{1}{\Delta x^2} - 1 + 3(u_2)^2\right) & -\frac{\Delta t}{\Delta x^2} & \dots & \dots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & -\frac{\Delta t}{\Delta x^2} & 1 + \Delta t(2\frac{1}{\Delta x^2} - 1 + 3(u_{N_x})^2) \end{pmatrix}$$

Newton's method for nonlinear problems (2/2)

Solve iteratively the linear systems

$$
\begin{aligned}
&J_{u^{n+1}} R(u^{(k)}) \delta u^{(k)} = -R(u^{(k)}) \\
&u^{(k+1)} = u^{(k)} + \delta u^{(k)}
\end{aligned}
$$

Starting from $u^{(0)} = u^n$.

Until convergence, i.e., $\|R(u^{(k)})\| < \varepsilon$ for some tolerance $\varepsilon > 0$ or until $\delta u^{(k)}$ is small enough, i.e., $\|\delta u^{(k)}\| < \varepsilon$.

2 Dimensions

2D Allen-Cahn equation

$$\partial_t u - (\partial_{xx} u + \partial_{yy} u) = u - u^3,$$

for $u : \Omega \times \mathbb{R}^+ \to \mathbb{R}$, with $\Omega \subset \mathbb{R}^2$ provided some

Initial conditions

$$u(x, t = 0) = u_0(x), \qquad x \in \Omega.$$

Boundary conditions

$$\begin{aligned} u(t, x) &= u_D(t, x), \qquad && t \in \mathbb{R}^+, x \in \Gamma_D \subset \partial\Omega, \\ \partial_x u(t, x) \cdot \mathbf{n} &= u_N(t, x), \qquad && t \in \mathbb{R}^+, x \in \Gamma_N \subset \partial\Omega. \end{aligned}$$

Semidiscretization in 2D on a rectangle $[x_{min}, x_{max}] \times [y_{min}, y_{max}]$

Take a grid of values, given by

$$x_{min} = x_0 < x_1 < \cdots < x_{N_x} = x_{max},$$
$$y_{min} = y_0 < y_1 < \cdots < y_{N_y} = y_{max}.$$

Then, we can look for an approximation of $u(t, x_i, y_j) \approx u_{i,j}(t)$ for all $i = 0, \ldots, N_x$ and $j = 0, \ldots, N_y$ that solves the following system of ODEs

$$\begin{cases} \frac{du_{i,j}}{dt} - \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2} - \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta y^2} = u_{i,j} - u_{i,j}^3, & \forall i = 1, \ldots, N_x - 1, j = 1, \ldots, N_y - 1, \\ \text{Initial condition} & \\ u_{i,j}(0) = u_0(x_i, y_j), & \forall i = 0, \ldots, N_x, j = 0, \ldots, N_y, \\ \text{Dirichlet BC e.g. left and right} & \\ u_{i,j}(t) = u_D(t, x_i, y_j), & \forall t \in \mathbb{R}^+, i = 0, N_x, j = 0, \ldots, N_y, \\ \text{Neumann BC e.g. bottom} & \\ \frac{u_{i,1}(t) - u_{i,0}(t)}{\Delta y} = u_N(t, x_i, y_0), & \forall t \in \mathbb{R}^+, i = 0, \ldots, N_x. \end{cases}$$

Vectorization

It's hard to deal systems of unknown matrices. We can vectorize the solution by stacking the columns of the matrix U into a vector $\mathbf{u}$ of size $N_x N_y$. MATLAB's choice:

$$\mathbf{u} = \begin{bmatrix} U(x_1, y_1) \\ U(x_1, y_2) \\ \vdots \\ U(x_1, y_{N_y}) \\ U(x_2, y_1) \\ \vdots \\ U(x_2, y_{N_y}) \\ \vdots \\ U(x_{N_x}, y_1) \\ \vdots \\ U(x_{N_x}, y_{N_y}) \end{bmatrix}, \qquad U = \begin{pmatrix} u_1 & u_{N_y+1} & \cdots & u_{N_y(N_x-1)+1} \\ u_2 & u_{N_y+2} & \cdots & u_{N_y(N_x-1)+2} \\ \vdots & \vdots & \ddots & \vdots \\ u_{N_y} & u_{N_y+N_y} & \cdots & u_{N_y N_x} \end{pmatrix}$$

Kronecker product

Given two matrices $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{p\times q}$, we define the Kronecker product of A and B as the matrix $A \otimes B \in \mathbb{R}^{mp\times nq}$ given by

$$A \otimes B = \begin{bmatrix} a_{11}B & \dots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \dots & a_{mn}B \end{bmatrix}.$$

To perform $\partial_{xx}u$ and $\partial_{yy}u$ we can use the Kronecker product to write the system in matrix form as

$$\frac{d\mathbf{u}}{dt} = \left(I_{N_y} \otimes D_x + D_y \otimes I_{N_x}\right)\mathbf{u} + F(\mathbf{u}),$$

$$D_x = \frac{1}{\Delta x^2}\begin{bmatrix} -2 & 1 & 0 & \dots & 0 \\ 1 & -2 & 1 & \dots & 0 \\ 0 & 1 & -2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & -2 \end{bmatrix}, \qquad D_y = \frac{1}{\Delta y^2}\begin{bmatrix} -2 & 1 & 0 & \dots & 0 \\ 1 & -2 & 1 & \dots & 0 \\ 0 & 1 & -2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & -2 \end{bmatrix}.$$

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kron(D_x,speye(N_y)) + kron(speye(N_x),D_y)
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